Shifting Attitudes Toward Vaccines During 2021 of the COVID-19 Pandemic

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Introduction

During the COVID-19 pandemic, people across the world were exposed to an overwhelming amount of information from government agencies, news outlets, and social media platforms, much of which was difficult to distinguish from misinformation, amongst conflicting messages from leaders (Ferreira Cacerese et al., 2022; Gisondi et al., 2022). Understanding how public perceptions of vaccines shifted during this time is important because it reveals how information environments influence attitudes toward previously well-supported public health measures. By analyzing social media content, this project aims to better understand how views on vaccines evolved throughout the pandemic and what these changes suggest about public responses to future health emergencies. Today, with declining trust in health institutions, changes to vaccine schedules, and a resurgence of preventable diseases, it is important to understand how public attitudes toward vaccines form and change over time (Howard, 2026; Jennings et al., 2023; Ledderer et al., 2026). Prior research on vaccine hesitancy and trust during the pandemic suggests that vaccine attitudes were shaped by uncertainty, public health messaging, and trust in institutions. This makes social media sentiment a useful area to study over time (Jennings et al.,2023; Ledderer et al., 2026). Because our dataset is limited mostly to tweets from 2021, our project focuses more specifically on how vaccine sentiment shifted on Twitter during that year of the pandemic.

Research Question and Hypothesis

Our research question asks: to what extent did vaccine-related sentiment and tweet volume change on Twitter during 2021 of the COVID-19 pandemic?

Our first hypothesis was that the proportion of negative sentiment toward vaccines would increase over time. We expected this because early optimism around vaccine rollout could have weakened as public debate over side effects, mandates, breakthrough cases, and new variants became more visible online.

Our second hypothesis was that the overall frequency of vaccine-related discussion would increase over time. We expected this because vaccines remained at the center of public conversation during 2021 through rollout, eligibility expansion, booster discussions, and the spread of new variants.

Research Procedure

We utilized a large, publicly accessible dataset of COVID-19 vaccine-related tweets from Kaggle. Because the data came from Twitter, our project measures vaccine-related discourse on that platform rather than public opinion in the population as a whole. In R, we processed and analyzed the data using the lubridate, janitor, and tidytext packages.

After importing the raw tweet data from the CSV file into R, we cleaned the column names and identified the main variables needed for the project. These included tweet ID, tweet text, date and time, language, follower count, retweet count, and available location information. To avoid changing the original data, we kept the raw tweet text and created a separate cleaned text version for analysis. We kept both the original tweet text and the cleaned tweet text so that we would not lose the original wording of the tweets. This also made it easier to compare the raw data to the processed version when checking whether the cleaning steps were working correctly. We converted the text to lowercase, removed links, usernames, punctuation, extra spaces, and other formatting characters that did not add meaning to the tweet. We removed hashtag symbols but kept the words themselves. We did this because a hashtag like #vaccine still contains useful meaning, even if the symbol itself does not help with the analysis. We also removed empty tweets that no longer contained usable text after cleaning. In addition, we standardized the date field and created month and year variables so that changes over time could be analyzed more easily.

We then filtered the dataset so that it focused only on tweets relevant to our research question. We kept tweets that matched vaccine-related keywords such as “vaccine,” “vaccination,” “booster,” “shot,” “Pfizer,” “Moderna,” and related COVID-19 terms. We restricted the dataset to English language tweets, and when location information was available, we also used that to help focus on U.S.-related tweets. To improve the quality of the data, we removed retweets, duplicate tweets, and obvious spam posts. Spam posts were identified using simple rules such as unusually high number of links, hashtags, or mentions, along with common promotional phrases. The final cleaned dataset included the raw and cleaned tweet text, date variables, follower count, retweet count, and location-related fields when they were available. The primary text variables were then cleaned tweets for sentiment analysis; the temporal variables were the date and month of the tweet; the engagement metric was the follower and retweet counts; the derived variables were the sentiment scores, sentiment category, and the tweet volume by month.

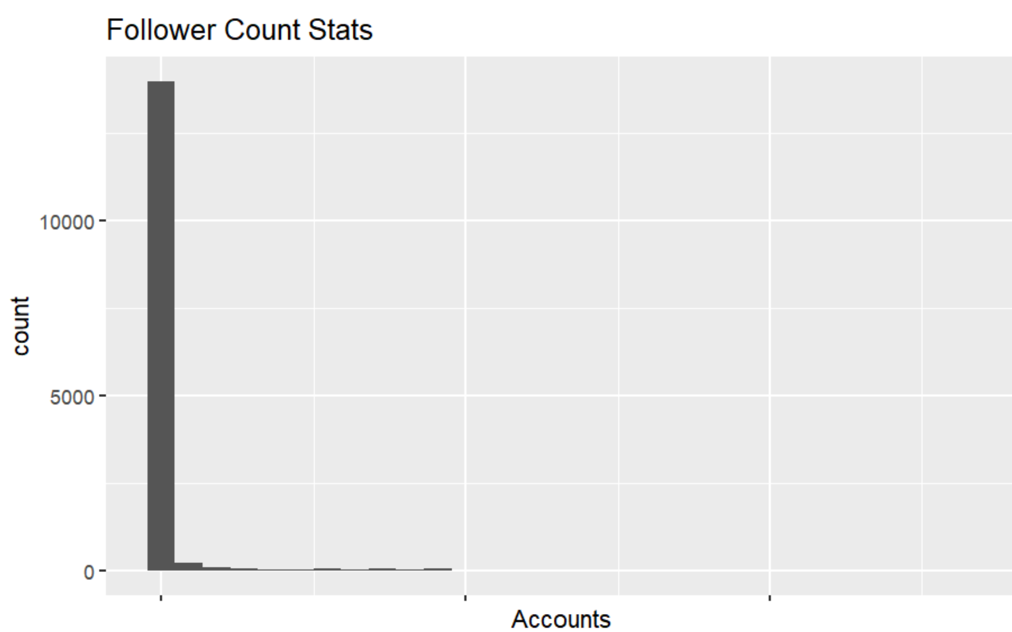
We spent time cleaning the tweets carefully because raw social media data is messy and can include repeated posts, links, formatting issues, and other text that does not reflect someone’s opinion. We wanted the final dataset to focus more on tweets that actually contained usable language about vaccines. At the same time, we recognize that text cleaning can remove some context, which means a negative word in a tweet may not always be directed at the vaccine itself.

For our analytical approach, we utilized descriptive statistics to summarize key features: monthly tweet volume assessed trends in vaccine-related discourse over time, and measures of central tendency compute the follower counts and retweet counts. The sentiment analysis was conducted using the AFINN lexicon, a dictionary that assigns integer sentiment scores to individual words ranging from -5 (most negative) to 5 (most positive). Since AFINN was developed as a general-purpose sentiment dictionary, it did not include much of the vaccine-specific language that appeared in our corpus. To address this, we created a separate list of vaccine and COVID-related terms in R and appended that list to the base AFINN dictionary before matching words to tweets. The added terms were grouped into categories such as vaccine efficacy and safety, adverse events and harm, misinformation and conspiracy, coercion and freedom, public health institutions, COVID-19 disease context, and emotional and social response. We assigned scores to these terms before running the final analysis based on how strongly positive or negative the words usually sounded in vaccine-related discussion. To try to match the patterns of the AFINN dictionary, we examined the scoring of terms in the base dictionary and assigned scores to our vaccine-related words through group discussion to include multiple perspectives on the scoring. This step involved some judgement, so it is possible that researcher bias influenced some of the added scores. For that reason, we treat the customized dictionary as a better fit for our topic, but not as a perfectly objective measure. For example, words like “immunized” and “lifesaving” were assigned positive scores, while terms like “hoax” and “unsafe” were assigned negative scores.

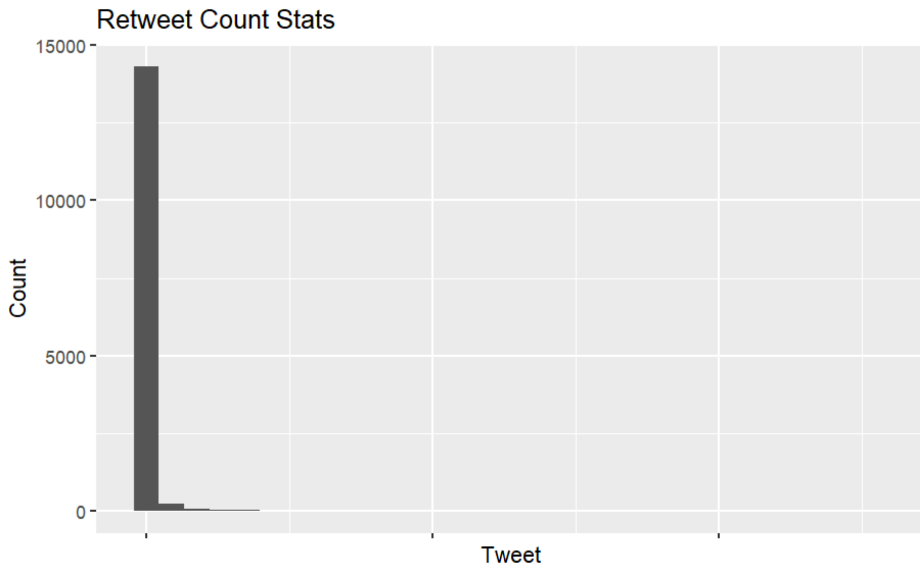
The custom dictionary was built by joining the group of vaccine-specific terms with the base AFINN dictionary. Tweets were then processed to account for multi-word phrases included in the custom list and tweets were then tokenized into individual words. Each token was then matched against the customized sentiment dictionary, and word-level scores were summed and averaged to produce sentiment scores for each tweet. Tweets were generally classified as positive, negative, or neutral based on whether their average score fell above, below, or at zero, respectively. A final temporal analysis was done by aggregating the average sentiment scores by month, and visualizations of distributions and scatterplots showed the relationship between follower count and sentiment.

The main statistical tests and models were simple linear regressions to examine the relationships between follower count, retweet count, and sentiment score with a fitted regression line. This was chosen to show where influence was associated with sentiment expression. We checked the assumptions visually by applying a log transformation to address the right skew of the follower count and retweet data points and to improve interpretability. Scatterplots were used to evaluate linearity and residual spread. Additionally, we used a chi-squared test to see if the distribution of negative sentiment differs across months. To do this we converted average sentiment by month to a categorical variable, positive or negative. This was part of our analysis of whether negative sentiment increased over time.

Graph 1: Skewness of Follower Counts



Graph 2: Skewness of Retweet Counts

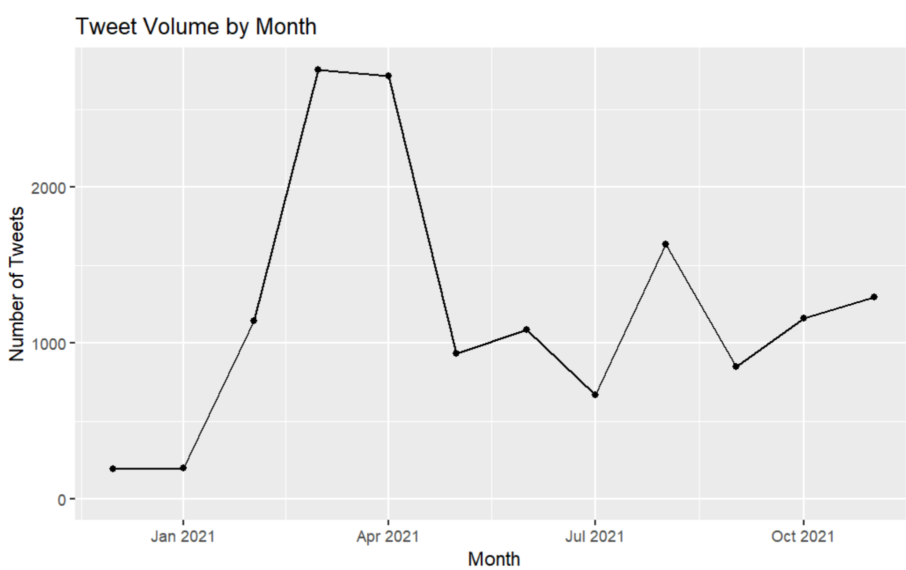


One limitation of the cleaning process was that some filters depended on the information available in the original dataset. For example, location-based filtering could only be used when usable location metadata was present, and spam detection was based on simple rule-based screening rather than manual review of every tweet.

Results

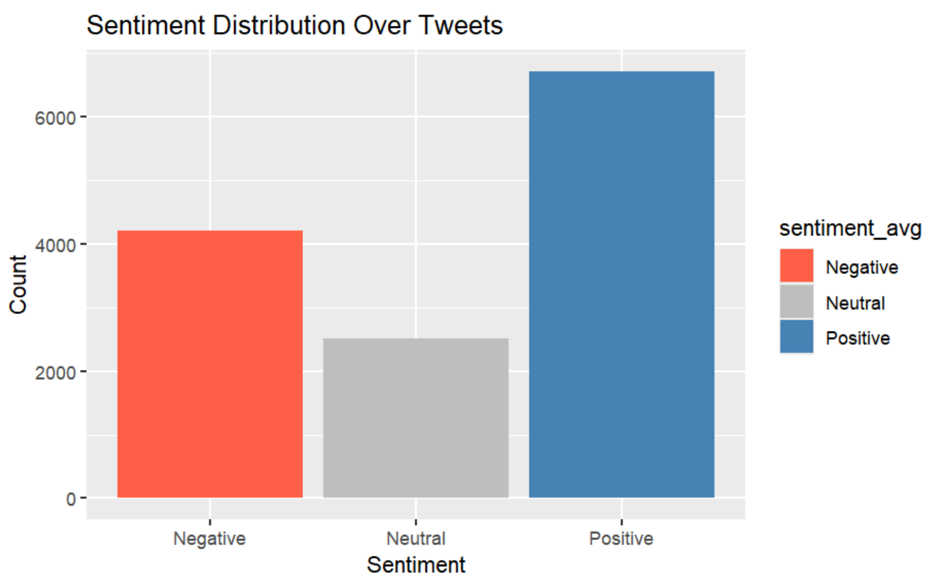
Before beginning the main analysis, we worked with a cleaned version of the tweet dataset rather than the raw file. After filtering and preprocessing, the final dataset contained 14,630 tweets. The cleaned file included the main variables used in the project, including raw tweet text, cleaned tweet text, date information, follower count, retweet count, and sentiment-related variables created later in the analysis.

Graph 3. Descriptive Statistics



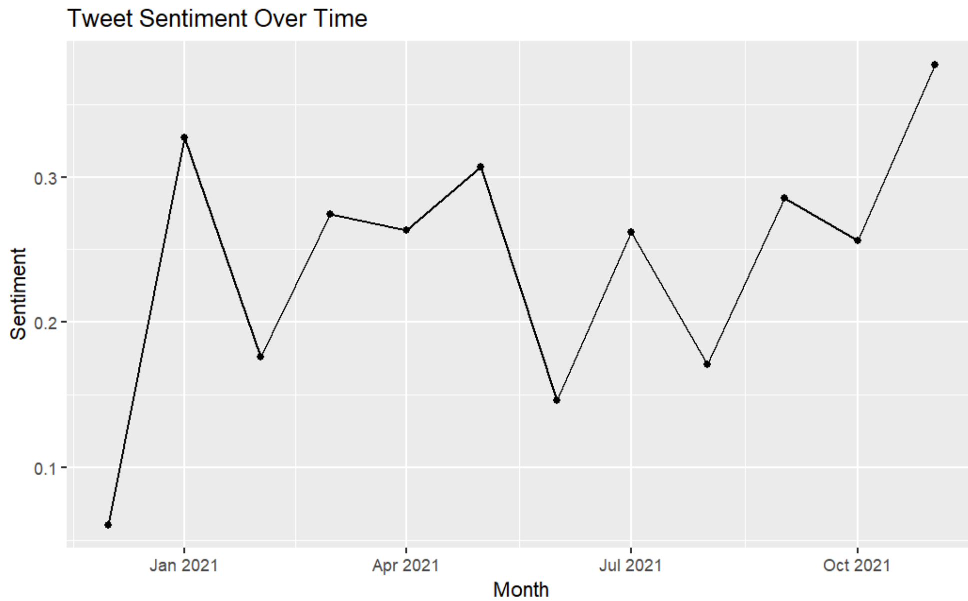
To test our hypothesis that the volume of tweets increased significantly over the course of the pandemic, we created descriptive statistics of our dataset. These descriptive statistics were calculated from the cleaned tweet dataset after filtering for vaccine-related content and removing unusable, duplicated, retweeted, and spam posts. This step helped us focus the analysis on tweets that were more likely to reflect original vaccine discourse rather than repeated or noisy content. We analyzed retweets, followers, and tweet volume. The retweet and follower counts were preliminary analyses used to summarize the data. We found that the average number of followers was 17,519, but the median was only 525. We also found the average number of retweets to be around 5, with a median of 0 and a maximum of 2550. These numbers are less important than the volume of tweets related to our hypothesis, but they are important later, once we conduct our regressions. When analyzing the volume of tweets in the data, we found a general upward trend interrupted by a sharp decline in May 2021, as seen in Graph 3. The peaks on Graph 3 indicate the months with the highest tweet volumes: March and April 2021. In March 2021, there were 2755 vaccine-related tweets, and in April 2021, there were 2714. There was a decline in tweets in May 2021, resulting in only 935 tweets. After May, there is a general upward trend, but the numbers never reach the same levels they did in March and April. While there was a decline in May, the overall trend in tweet volume supports our second hypothesis: tweet volume increased over the course of the pandemic. This is also consistent with the stage the pandemic was in during those months.

Graph 4. Positive, Negative, Neutral Sentiment



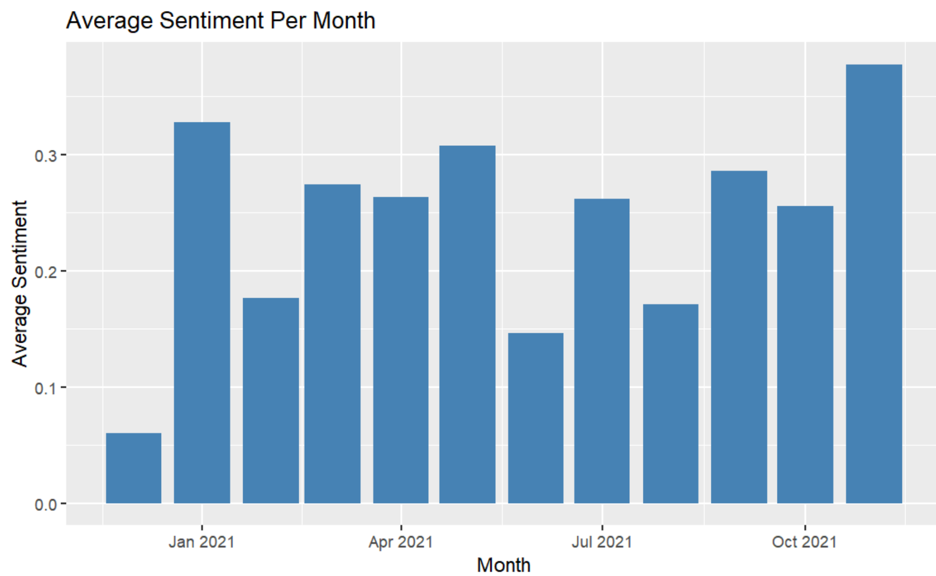
While conducting our sentiment analysis of the data, we also examined the overall distribution of tweet sentiment to determine which sentiment was most common. We found that 31.4% of the tweets had a negative sentiment. This initial analysis made us question the strength of our first hypothesis that negative sentiment would increase over time. However, negative sentiment could increase for the duration of our observation period and still not be the prevailing sentiment. We also found that 18.9% of the tweets have a neutral sentiment. Thus, a small proportion of our tweets are neither in favor of nor against vaccines, which is normal when assessing public perception. Finally, we found that 49.9% of the tweets have a positive sentiment. This evidence helps visualize how tweet sentiment spread over the observation period, which was important for analyzing our hypotheses.

Graph 5. Average Sentiment Over Time



Once we completed the analysis of overall tweet sentiment, we examined the average sentiment over time to test our hypothesis. Consistent with the results of our first analysis, we found evidence that the majority of sentiment was positive. As shown in Graph 4, the tweet sentiment gets more positive over time. Starting in December 2020, we see a sharp increase in tweet sentiment through January 2021. Between January and February, tweet sentiment declines, but the effect is short-lived. Once we reach June, we observe a decline that remains below June levels until October. Finally, at the end of the year, we see another increase in sentiment to the end of 2021. The variation in the graph indicates that there were declining perceptions of vaccines over time, but that trend was not consistent throughout the year, and sentiment ended upward. Overall, the graph suggests that vaccine sentiment remained more positive than negative for much of the year, but it did not move in a single, steady direction from month to month. Instead, the uneven pattern suggests that sentiment may have responded to specific events during 2021, which would be a useful area for future analysis.

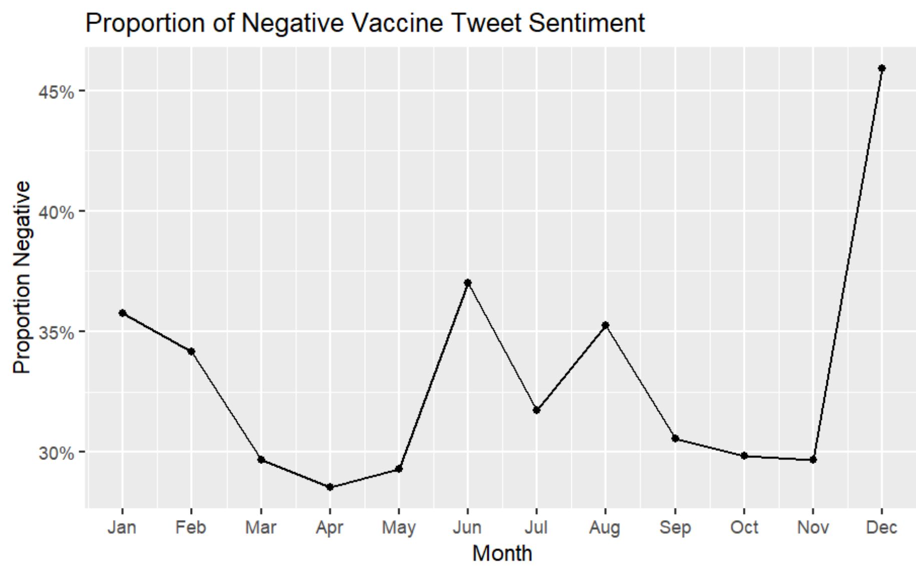
Graph 6. Average Sentiment by Month



Along with tweet sentiment over time, we examined how it changed from month to month. This analysis gave us a clearer picture of when public perception of vaccines changed, which helps inform our conclusions. Graph 5 shows the average sentiment for each month starting in December 2020. The first thing we noticed when analyzing the data is that all of the months have positive sentiment. This is consistent with the earlier statistic showing that 49.9% of sentiments were positive. There are negative sentiments, but they are overshadowed by the positive ones. The decline observed in the overall time analysis is also evident in this graph. In June and August, there was a decline in public perception about vaccines. This analysis shows us the exact months the decline was greatest, June and August, which helps better inform our overall conclusions and further research questions. Additionally, this analysis helped us determine which months had the highest sentiment: January and November of 2021, and the lowest sentiment: December 2020 and July 2021. This pattern suggests that the strongest positive months may have aligned with early optimism surrounding vaccine rollout, while lower months may have reflected greater uncertainty during the spread of new variants (CDC, 2024). These explanations are plausible, but they should be treated as interpretations rather than as direct proof of causation.

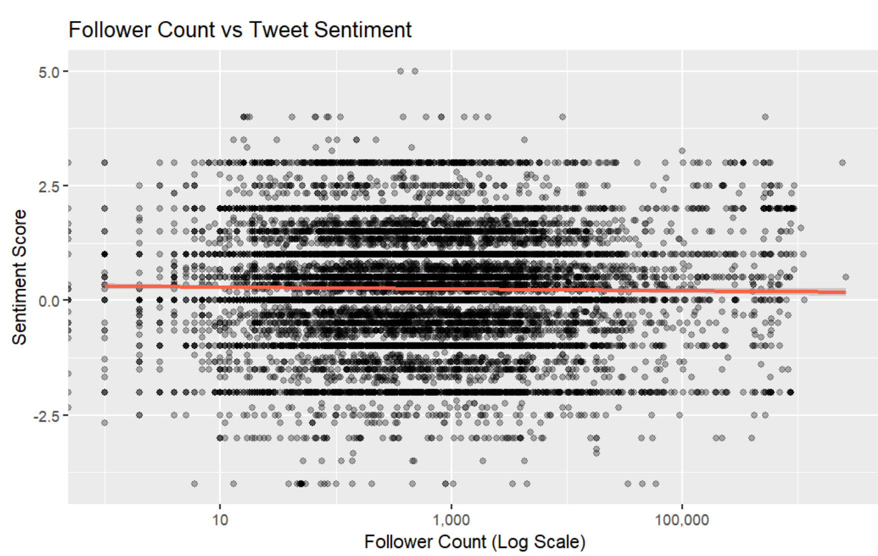
Using a chi-squared test to delve deeper into our analysis of negative sentiment, we found a significant relationship between negative sentiment and month, $\chi2$ (22, N=13,434) = 120.07, $p<.001$. Therefore, we have evidence that negative sentiment varied greatly across months during our observation period. This evidence is important to our analysis because, if there were no variation in negative sentiment by month, our hypothesis of increasing sentiment over the observation period would have to be rejected.

Graph 7: Proportion of Negative Sentiment by Month



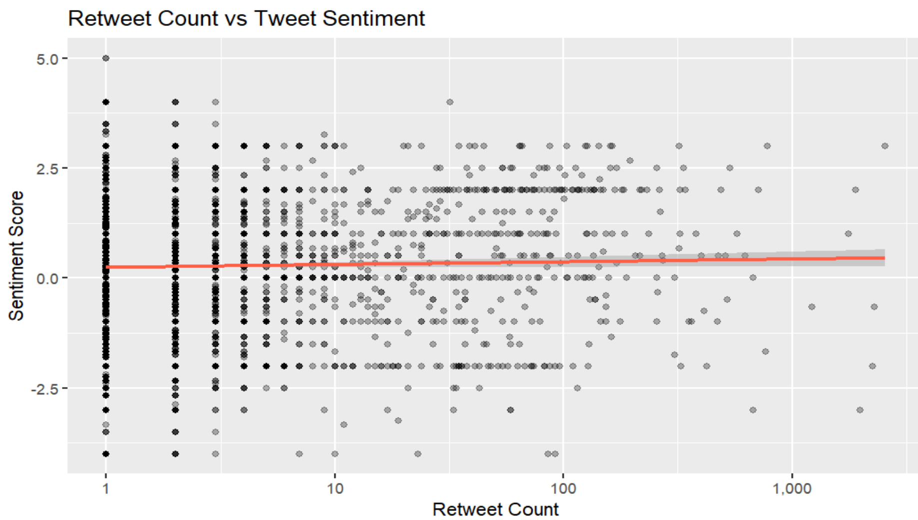
We also graphed the monthly proportion of tweets classified as negative to examine our first hypothesis more directly. Unlike the overall sentiment average, this measure focuses specifically on how common negative tweets were in each month. The proportion of negative tweets did not show a steady month-to-month rise across 2021. Instead, it fell during early spring, rose again in June and August, and then declined before increasing sharply in December. This pattern suggests that negative vaccine sentiment fluctuated over time rather than increasing in a simple linear way, which is why our first hypothesis was not supported.

Graph 8. Correlation Between Sentiment and Followers



As an additional analysis, we wanted to determine if there was a correlation between follower count and the sentiment score of tweets. Given the nature of follower counts and the high likelihood that they would skew the data, we used the log of follower counts to address this issue. Upon plotting these two data points on a scatterplot, we found no correlation. As shown above, the follower count does not show a clear trend in the sentiment score. Additionally, because the regression line is flat, we know there is virtually no correlation between these two variables. Therefore, we concluded that there is no correlation between a person's follower count and whether a tweet is positive or negative towards vaccines. Another feature of this graph that supports our conclusion is the data's variance. Based on the graph, the variance of the data is roughly equal across follower count. Therefore, the data are homoskedastic, indicating that the lack of correlation holds consistently across all accounts.

Graph 9. Correlation Between Retweets and Sentiment



As the final part of our analysis, we analyzed whether there was a correlation between the number of retweets and the tweet sentiment. As in the previous regression, because of the variance in the retweet data, we took the log of this variable to avoid introducing bias into the model. There is no meaningful correlation between retweet count and tweet sentiment. The regression line is relatively flat, similar to the follower count correlation. The data points are relatively spread out across sentiment values, so whether tweets are retweeted does not depend on their positivity or negativity. There is a slight upward trend in the data when the retweet count is very high, which is consistent with people resharing helpful or positive tweets to help others feel better during the pandemic. However, this trend is very small. There is a cluster of retweets near the lower end of the spectrum, which is normal on Twitter and was observed in our initial descriptive statistics analysis.

Conclusions

Our results led us to reject the first hypothesis and only partially support the second. We originally expected negative sentiment to increase over time because vaccine discussion became more politicized during 2021, especially around mandates, side effects, breakthrough cases, and new variants. However, our data did not show a steady rise in negativity. Instead, positive tweets were the most common category overall, and the time-based graphs showed fluctuation rather than a simple downward shift in sentiment. This suggests that our first hypothesis may have been too linear for a topic that changed in response to specific events throughout the year.

The second hypothesis received partial support. Tweet volume clearly rose from the beginning of our observation period and peaked in March and April 2021, but it did not continue to rise steadily after that point. Because of the sharp drop after April, it is more accurate to say that vaccine discussion intensified in early 2021 and then remained active at a lower level rather than increasing continuously across the full year.

Our additional analyses found no notable correlation between sentiment and either follower count or retweet count, suggesting that the positivity or negativity of a tweet was unrelated to the size or reach of the account posting it. This finding indicates that vaccine sentiment on Twitter during this period was distributed across different types of users rather than being concentrated among influential accounts.

The assumption that vaccine sentiment deteriorated over the course of the pandemic is not supported by this set of Twitter data from 2021. However, these findings should be interpreted carefully. Twitter users may not be representative of the general public, and positive sentiment in tweets does not necessarily reflect actual vaccination behavior. In addition, because our dataset is limited mostly to 2021, our conclusions apply more directly to that year than to the full pandemic period. Future research could test whether these same patterns appear across a longer time span or on other social media platforms.

General AI Statement

AI was used to help supplement the list of vaccine-related terms we had come up with. We used this to add as many relevant terms to the AFINN dictionary as possible, so that it would be sufficiently tailored to our topic and yield more accurate results. Link to chat: https://claude.ai/share/7ceb5f9e-ee21-4643-ac42-f1f30d2eaaf2 (Note: the first prompt isn't showing up because it was written in a separate document and then pasted in. The text of this prompt is included in an appendix.) The output of this chat, including the scores assigned by AI, was then reviewed, revised, and used to help with the refining of the AFINN dictionary.

Contributions Statement

- Sarah: I researched potential datasets, determined the procedure and wrote the code for adding terms to the AFINN dictionary and processing the tweets to find sentiment values for the first part of the sentiment analysis, and contributed to the research procedure, results, and conclusion sections of the report.
- Austin: I researched potential datasets, wrote the code for the second part of the sentiment analysis and correlation visualizations, drafted the results section, and contributed to the conclusions and research question/hypothesis sections.
- Annabelle: I wrote all of the code regarding descriptive statistics and drafted the introduction, research question and hypothesis, and research procedure sections of the written report.
- Seth: I imported and cleaned the tweet dataset in R and did the filtering pipeline for vaccine-related tweets. I then made adjustments to the cleaned dataset to make sure follower count and number of retweets were kept in the final cleaned dataset. I also contributed to the research procedure and results section of the paper.

References

CDC. (2024, July 8). CDC museum COVID-19 timeline. Centers for Disease Control and Prevention. https://www.cdc.gov/museum/timeline/covid19.html#Early-2021

Ferreira Caceres, M. M., Sosa, J. P., Lawrence, J. A., Sestacovschi, C., Tidd-Johnson, A., Rasool, M. H. U., Gadamidi, V. K., Ozair, S., Pandav, K., Cuevas-Lou, C., Parrish, M., Rodriguez, I., Fernandez, J. P. (2022). The impact of misinformation on the COVID-19 pandemic. *AIMS Public Health*, *9*(2), 262–277. https://doi.org/10.3934/publichealth.2022018

Gisondi, M. A., Barber, R., Faust, J. S., Raja, A., Strehlow, M. C., Westafer, L. M., & Gottlieb, M. (2022). A Deadly Infodemic: Social Media and the Power of COVID-19 Misinformation. *Journal of Medical Internet Research*, *24*(2), e35552. https://doi.org/10.2196/35552

Howard, B. (2026, March 3). What you need to know about the updated childhood vaccination schedule. *AAMCNews*. https://www.aamc.org/news/what-you-need-know-about-updated-childhood-vaccination-schedule

Jennings, W., Valgarðsson, V., McKay, L., Stoker, G., Mello, E., & Baniamin, H. M. (2023). Trust and vaccine hesitancy during the COVID-19 pandemic: A cross-national analysis. *Vaccine: X*, *14*, 100299. https://doi.org/10.1016/j.jvacx.2023.100299

Ledderer, L., Nielsen, K. H., Skodborg, L., & Fage-Butler, A. (2026). Public trust and mistrust of COVID-19 vaccines: A systematic meta-narrative review. *Vaccine*, *69*, 127947. https://doi.org/10.1016/j.vaccine.2025.127947

Verywell Health. (2024, September 25). A Timeline of COVID-19 Variants. Verywell Health. https://www.verywellhealth.com/covid-variants-timeline-6741198

Appendix

Initial Prompt Used for Term Search

I am working on a project where I am trying to analyze the sentiments of a dataset of tweets to determine sentiments about vaccines during the covid pandemic. Here is the intro and hypothesis for additional context: "Introduction

During the COVID-19 pandemic, people across the world were exposed to an overwhelming amount of information from government agencies, news outlets, and social media platforms, much of which was difficult to distinguish from misinformation, amongst conflicting messages from leaders (Ferreira Cacerese et al., 2022; Gisondi et al., 2022). Understanding how public perceptions of vaccines shifted during this time is important because it reveals how information environments influence attitudes toward previously well-supported public health measures. By analyzing social media content, this project aims to better understand how views on vaccines evolved throughout the pandemic and what these changes suggest about public responses to future health emergencies. Today, with declining trust in health institutions, changes to vaccine schedules, and a resurgence of preventable diseases, it is critical to understand how public attitudes toward vaccines form and change over time (Howard, 2026; Jennings et al., 2023; Ledderer et al., 2026). Therefore, our team wants to know to what extent public perceptions of vaccines change during the COVID-19 pandemic.

Research Question and Hypothesis

Our research question asks: to what extent do public perceptions of vaccines change during the COVID-19 pandemic? First, we hypothesize that the proportion of negative sentiment towards vaccines increased as the COVID-19 pandemic progressed. If this hypothesis is true, we expect to see a higher percentage of negative comments and posts about vaccines in the media and social media content as time passes. Second, we hypothesize that the overall frequency of vaccine-related discussions increased significantly over the course of the COVID-19 pandemic. If this hypothesis is true, we expect to see a measurable increase in the number of vaccine-related posts and social media interactions over time."

I want to use the AFINN sentiment dictionary and I want to customize the dictionary by adding vaccine/covid specific words with corresponding scores. Here are some I've come up with in the format that I'm going to put them in my code, can you help me come up with more terms within these categories to add to AFINN, for two-word phrases, try to minimize these, but you can put an underscore between words:

```
vaccine_terms <- tribble(
  ~word,              ~value,
# Vaccine efficacy & safety
  "immunized",            4,
```

```
"vaccinated",            3,
"protection",            3,
"effective",             3,
"safe",                  3,
"immunity",              3,
"herd_immunity",         3,
"life_saving",           5,
"lifesaving",            5,

# Adverse events & harm
"side_effect",          -2,
"side_effects",         -2,
"adverse",              -2,
"adverse_event",        -3,
"adverse_reaction",     -3,
"myocarditis",          -3,
"blood_clot",           -3,
 "injury",              -3,
"injured",              -3,
"toxic",                -4,
"unsafe",               -4,

# Misinformation & conspiracy
"conspiracy",           -3,
"hoax",                 -4,
"propaganda",           -3,
"scam",                 -4,
"corrupt",              -4,
 "agenda",              -2,
"microchip",            -2,
"experimental",         -2,
"untested",             -3,

# Coercion & freedom
"mandate",              -1,
"mandatory",            -2,
"forced",               -3,
"coercion",             -4,
"discrimination",       -4,
"coerce",               -4,
```

```
  # Public health & institutions
  "skeptical",          -2,
  "hesitant",           -2,
  "confident",           3,
  "transparent",         3,
  "accountability",      2,
  "credible",            3,

  # COVID-19 disease context
  "covid",              -2,
  "coronavirus",        -2,
  "pandemic",           -3,
 "contagious",          -2,
 "long_covid",          -4,
 "quarantine",          -2,
  "lockdown",           -3,
  "isolation",          -2,

  # Emotional & social response
   "relieved",           3,
  "gratitude",           4,
  "hopeful",             3,
  "optimistic",          3,
   "anxious",           -2,
  "fearful",            -3,
  "scared",             -3,
  "worried",            -2,
  "concerned",          -1,
  "furious",            -4,
  "selfish",            -3,
  "selfless",            3,
  "sacrifice",           2
)
```